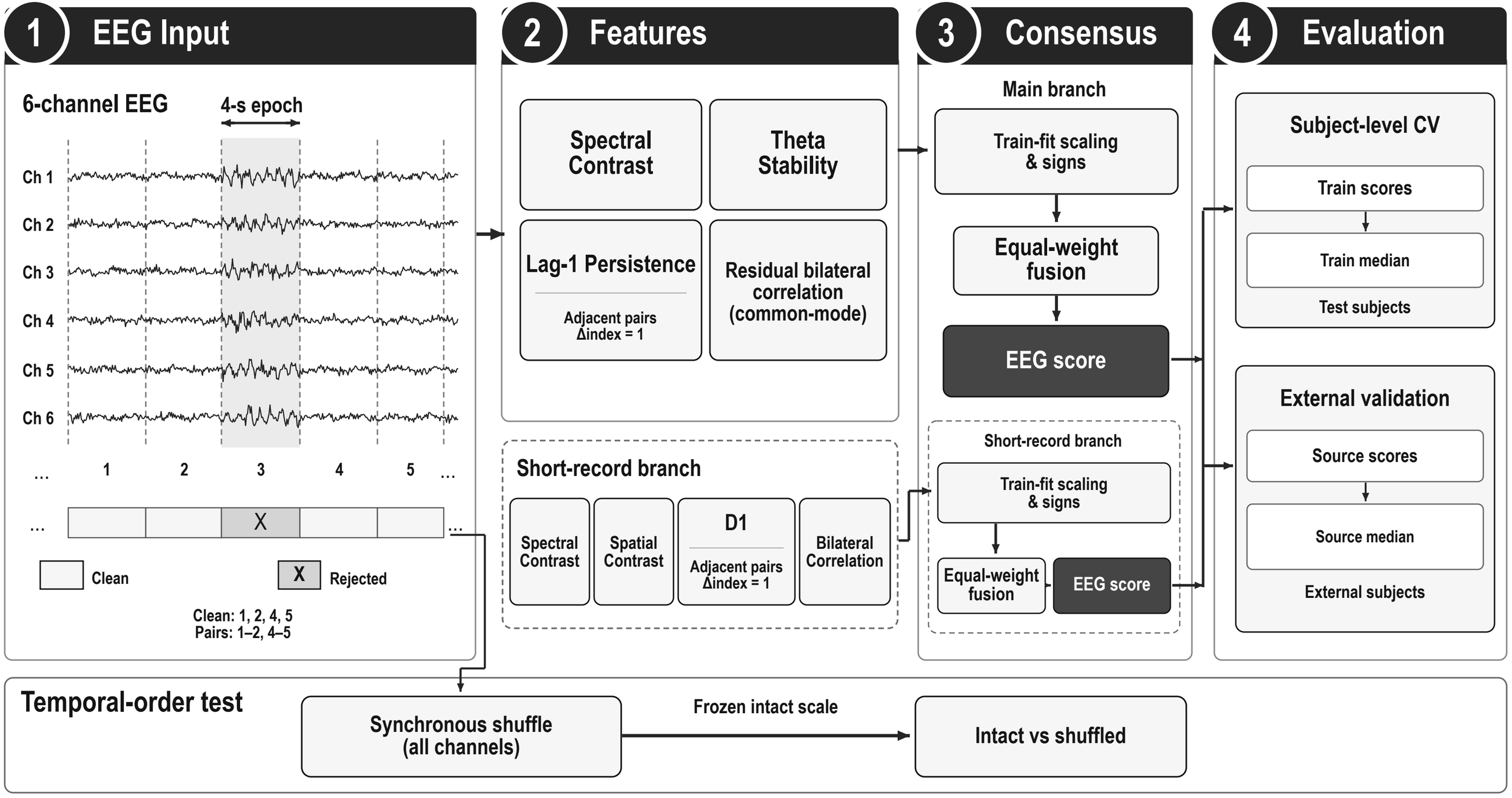

1
EEG Input
6-channel EEG
4-s epoch
Ch 1
Ch 2
Ch 3
Ch 4
Ch 5
Ch 6
… 1 2 3 4 5 …
… X …
Clean
X
Rejected
Clean: 1, 2, 4, 5
Pairs: 1–2, 4–5
2
Features
Spectral Contrast
Theta Stability
Lag-1 Persistence
Adjacent pairs
Δindex = 1
Residual bilateral correlation (common-mode)
Short-record branch
Spectral Contrast
Spatial Contrast
D1
Adjacent pairs
Δindex = 1
Bilateral Correlation
3
Consensus
Main branch
Train-fit scaling & signs
Equal-weight fusion
EEG score
Short-record branch
Train-fit scaling & signs
Equal-weight fusion
EEG score
4
Evaluation
Subject-level CV
Train scores
Train median
Test subjects
External validation
Source scores
Source median
External subjects
Temporal-order test
Synchronous shuffle (all channels)
Frozen intact scale
Intact vs shuffled